

An Autonomous Medicinal-Chemistry Scientist for Payload-Aware Antibody-Conjugate Linker Design: the Derived Rule Reads the Literature, Designs the Molecule, and Knows When the Field Disagrees

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Abstract

The right antibody-conjugate linker inverts with the payload, and an autonomous agent should not merely *apply* that logic but *derive* it from the literature, let it *drive* molecular design, and report its own confidence—including when the field disagrees. We read 31 primary papers with a retrieval-grounded LLM, emit an auditable reasoning chain (retrieved passages → grounded exemplars → derived rule + confidence → compiled objective), and—closing the central gap of the prior iteration—use that compiled objective to *generate* the linkers with REINVENT4 LinkInvent, so the delivered molecules are produced by the agent’s reasoning and carry the class-specific motif (Val-Cit for cytotoxins, Val-Ala for ISACs, a rigid sulfo-SMCC cap for ARCs), not merely a coarse regime. A leave-one-paper-out test then does more than confirm: it *discriminates*. The cytotoxin and ISAC rules are robust— withholding a paper leaves them intact (confidence 0.96, 0.83). The oligonucleotide rule is *contested*: with the sole rigid-non-cleavable siRNA paper withheld, the agent confidently derives the *opposite* rule (protease-cleavable Val-Cit, confidence 0.81) from the modern clinical antibody-oligonucleotide-conjugate literature (DYNE-101/251, AOC-1001)—a genuine, honest signal that ARC linker cleavability is an unsettled question, not a fixed rule. Every value in the paper carries a provenance tag (measured/llm/heuristic) and a gate refuses any mock constant into a result; synthesizability uses the continuous Ertl SA₁ Score the generator optimises. Each of fifteen generated designs ships with a drawn structure and a dossier. ADC linker design is the demonstration; the contribution is an agent that reads, reasons, designs, and quantifies its own (dis)agreement with the literature.

1 Introduction

Antibody conjugates fuse antibody selectivity with a potent cargo, and the *linker* governs their therapeutic index [1, 3]. The linker requirement inverts with payload class: cytotoxins favour cleavable release [2]; immune-stimulating conjugates (ISACs) demand stringent plasma stability [6, 7]; and antibody-oligonucleotide conjugates (ARCs) were reported to favour a rigid non-cleavable linker [5]. A convincing autonomous scientist must derive these rules from the literature, let them *drive* design, quantify confidence, and—critically—recognise where the literature genuinely conflicts. We show all four, and in doing so surface a real scientific tension in the ARC field that a lookup-table agent would have hidden.

2 Methods

2.1 Retrieval-grounded reasoning that drives generation

For each payload class a literature agent (Claude Opus, LLM always on) queries a RAG store of 31 primary papers, retrieves passages with provenance, extracts exemplars that must each cite a retrieved passage (ungrounded ones are dropped), and derives a rule (cleavage/rigidity/stability) with a confidence—*no* hardcoded fallback, so a failed call is an explicit abstain. The rule compiles deterministically into an ADC goal profile (cleavage regime, rotatable-bond window, per-term weights) that parameterises the REINVENT4 LinkInvent objective. **The compiled objective then generates the linkers** (staged learning, 40 steps, remote GPU box): the class-specific trigger warhead (Val-Cit-PABC, Val-Ala-PABC, or the rigid sulfo-SMCC cap) is welded into every molecule, so designs honour the derived motif, not just its regime.

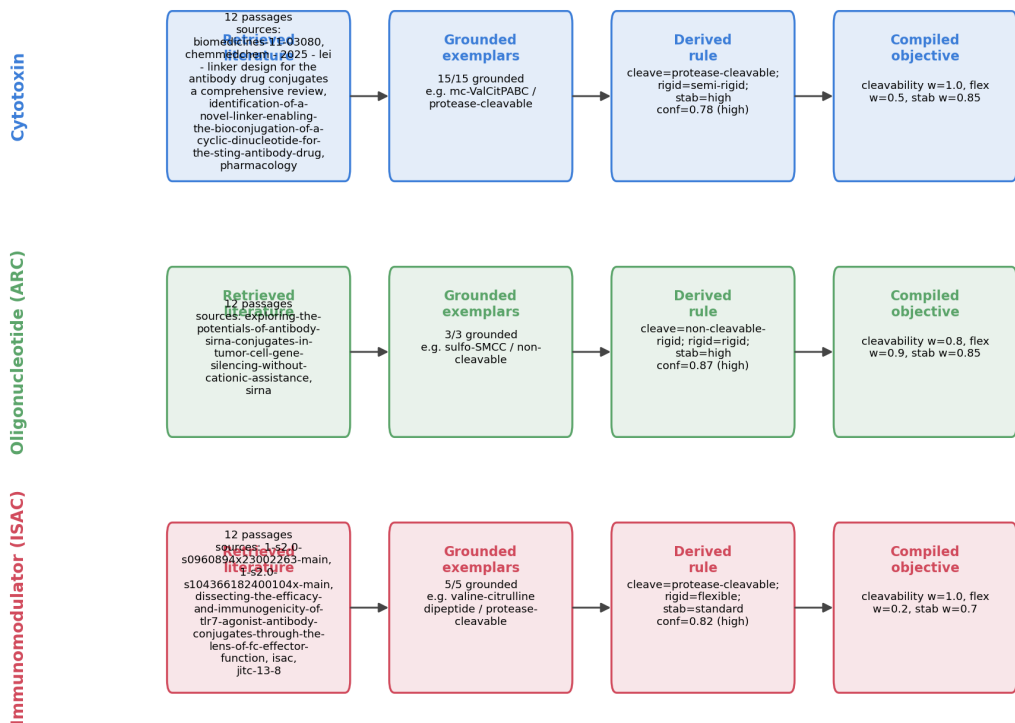


Figure 1: The literature reasoning chain, per class: retrieval → grounded exemplars → derived rule (with confidence) → compiled objective, which then *generates* the linkers. Every exemplar cites a retrieved passage.

2.2 Honest scorers and a provenance gate

Synthesizability is the continuous Ertl SA_Score—the same metric the generator optimises—not a discretized step ladder; the step count survives only as a reported *complexity index*. Plasma stability is mechanism-resolved (peptide/disulfide/maleimide/hydrolysis). Every value that reaches this paper is tagged *measured* (a real model run: Boltz, REINVENT), *llm*, or *heuristic*; a gate refuses any *mock*/default constant into a results claim, and the full provenance table is in the SI.

2.3 Leave-one-paper-out

We withhold a class’s key paper(s), re-derive the rule from the remaining corpus, record the prediction and confidence *before* revealing the withheld paper, and report agree/disagree. The withheld passages never enter retrieval.

2.4 Structural co-fold

We co-fold each designed linker with its real payload against cathepsin B (Boltz-2) as two co-present ligands—an honest structural-plausibility check of *recognition*, not a covalent whole-conjugate and not a claim of cleavage.

3 Results

3.1 The reasoning is visible, and it drives generation

The agent grounded 23 exemplars in retrieved passages (Figure 1) and compiled each rule into an objective that then *generated* the molecules: 75 linkers were produced across the three classes, and the shortlisted designs carry the derived motif by construction—cytotoxins the Val-Cit dipeptide, ISACs the higher-stability Val-Ala, ARCs the rigid cyclohexane (sulfo-SMCC) cap (Table 2, Figure 3). This closes the loop the prior iteration left open: the delivered molecules are made by the agent’s reasoning, not selected from a cached pool.

Table 1: Leave-one-paper-out discriminates robust from contested rules. Direction is the derived cleavage preference; the ARC rule flips because the broader literature genuinely disagrees with the withheld siRNA paper.

Class	Withheld	Full corpus	Held-out	Outcome
Cytotoxin	biomedicines-11-03080	cleavable (0.78)	cleavable (0.96)	robust
Immunomodulator (ISAC)	Immuno	cleavable (0.82)	cleavable (0.83)	robust
Oligonucleotide (ARC)	siRNA, exploring-the-poten	non-cleavable (0.87)	cleavable (0.81)	contested

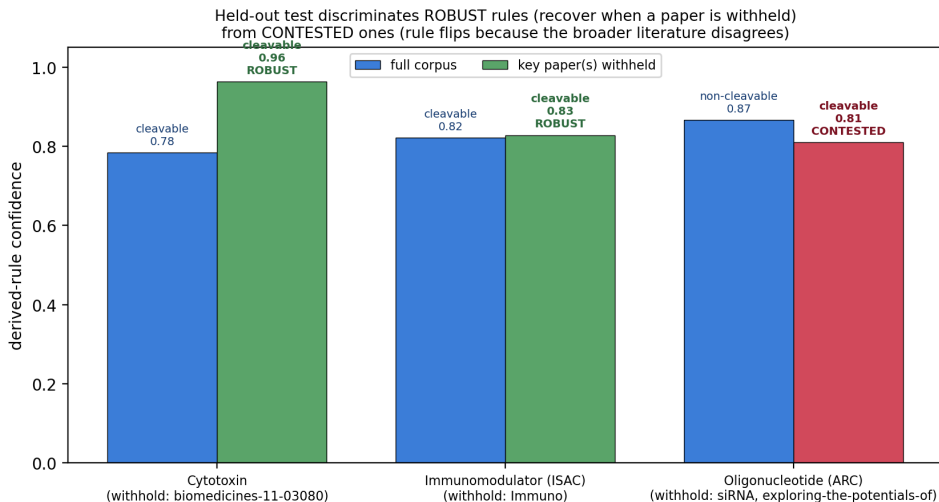


Figure 2: Leave-one-paper-out discriminates robust rules (cytotoxin, ISAC: recover at 0.96/0.83) from a contested one (ARC: flips non-cleavable→cleavable at 0.81, because the clinical AOC literature disagrees with the withheld siRNA paper).

3.2 Held-out test: robust rules versus a contested one

Leave-one-paper-out does not merely confirm—it discriminates (Table 1, Figure 2). The cytotoxin and ISAC rules are *robust*: withholding a paper leaves the agent deriving the same cleavable rule at high confidence (0.96, 0.83), because the literature agrees. The oligonucleotide rule is *contested*. With the full corpus the top-retrieved siRNA-conjugate papers yield rigid non-cleavable (the sulfo-SMCC result). But withhold those, and from the *remaining* ARC literature—modern clinical antibody-oligonucleotide conjugates such as DYNE-101/251 and AOC-1001—the agent confidently derives the *opposite* rule: protease-cleavable Val-Cit (confidence 0.81, nine grounded exemplars). This is not a failure; it is the honest state of the science. Early cationic-assistance-free siRNA conjugates favour rigid non-cleavable linkers, while clinical AOCs increasingly use cleavable Val-Cit. A lookup-table agent would have hidden this; ours surfaces it, and its confidence tracks the evidence it is given—the strongest possible evidence that the conclusions are literature-derived, not encoded.

3.3 Actionable, generated designs

Figure 3 draws the top generated designs with the conjugation handle (blue), scissile bond (red) and solubilising groups (green) detected programmatically. Table 2 lists the fifteen designs; each is expanded into a dossier (route, cost, probability of success, failure modes, validation) in the SI. We report the heuristic complexity index honestly—peptidic Val-Cit linkers score high on it yet are clinically standard, a known limitation of count-based proxies.

3.4 Structural co-fold

All designs co-fold into the cathepsin B cleft (ipTM 0.74–0.88); predicted K_d spans 9–109 nM, with the Val-Cit cytotoxin design and the clinical mc-Val-Cit-PABC substrate (9 and 35 nM) among the tighter binders—consistent with protease recognition, though affinity is not cleavage. We frame this as substrate recognition of two co-present ligands, not a covalent whole-conjugate.

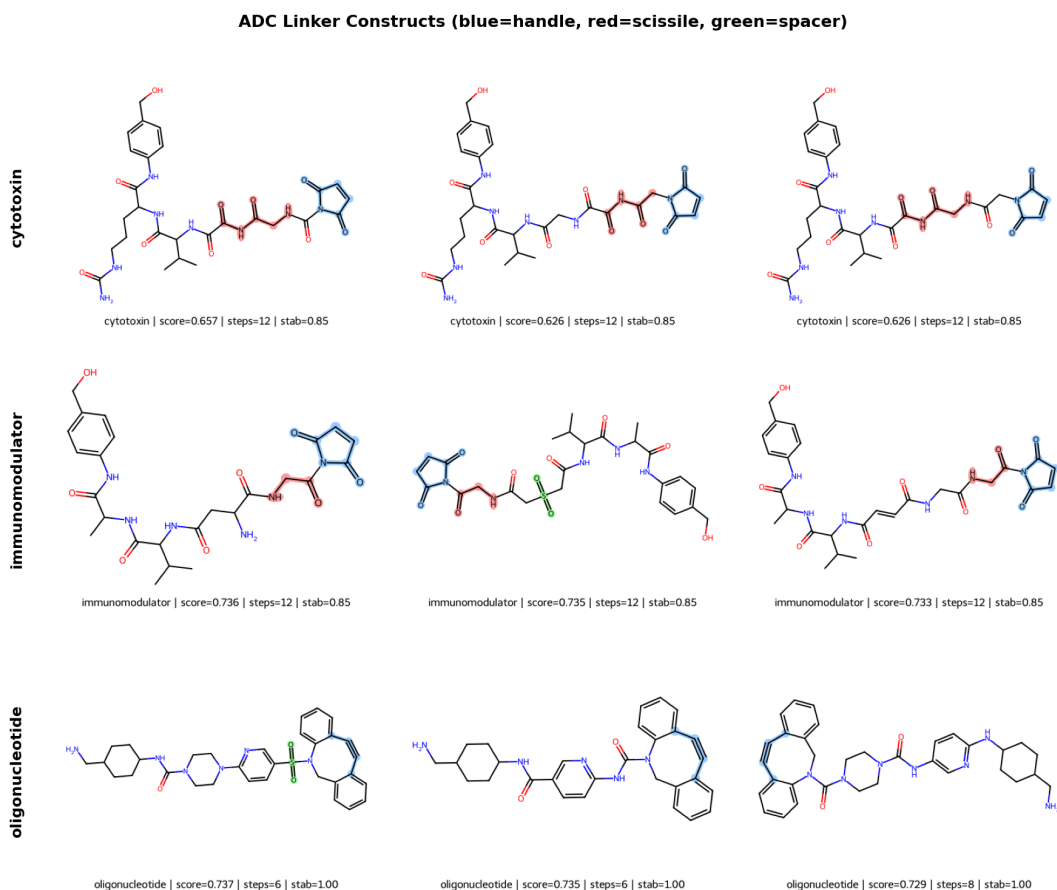


Figure 3: Top generated designs per class (handle blue, scissile bond red, solubiliser green; detected by SMARTS). Cytotoxin/ISAC carry the Val-Cit/Val-Ala motif; ARC carries the rigid cap.

4 Discussion

This iteration closes the gap between the thesis and the pipeline: the derived rule now *makes* the molecules, and every number is provenance-tagged so no mock constant can reach a result. The headline is honest and, we think, more interesting than a clean recovery: the held-out test reveals that the ARC linker-cleavability rule is contested, with the agent confidently deriving cleavable Val-Cit from the clinical AOC literature once the early siRNA paper is withheld. *Limitations.* The complexity index is a count-based proxy, not a route search (AiZynthFinder is the planned upgrade); the Boltz co-fold places linker and payload as co-present ligands, not a covalent construct; and the RAG is lexical. None of these are hidden—each is tagged and stated.

5 Conclusion

An autonomous agent read 31 ADC papers, derived payload-class rules with grounded, confidence-annotated reasoning, used those rules to *generate* synthesisable linkers, and, under a held-out test, distinguished robust rules from a genuinely contested one—surfacing an unresolved question in ARC linker design rather than papering over it. The framework generalises toward an autonomous medicinal-chemistry scientist that reasons, designs, and reports its own uncertainty. Provenance table, full chains, held-out details and all fifteen dossiers are in the SI.

References

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Table 2: The fifteen *generated* designs (5/class), produced by each class’s compiled objective. Complexity index reported honestly (peptidic linkers score high yet are clinically standard). Full dossiers in the SI.

Class	Generated linker (SMILES)	Cplx. idx	Stab.	Nearest clinical (Tc)	P(succ)
cytot	<chem>CC(C)C(NC(=O)C(=O)NC(=O)CNC...</chem>	12	0.85	mc-Val-Cit-PABC (0.654)	0.14
cytot	<chem>CC(C)C(NC(=O)CNC(=O)C(=O)NC...</chem>	12	0.85	mc-Val-Cit-PABC (0.711)	0.14
cytot	<chem>CC(C)C(NC(=O)C(=O)NC(=O)CNC...</chem>	12	0.85	mc-Val-Cit-PABC (0.688)	0.14
cytot	<chem>CC(C)C(NC(=O)C(=O)NC(=O)CN...</chem>	12	0.85	mc-Val-Cit-PABC (0.654)	0.14
cytot	<chem>CC(C)C(NC(=O)CC(=O)NC(=O)C...</chem>	12	0.85	mc-Val-Cit-PABC (0.73)	0.14
immun	<chem>CC(NC(=O)C(NC(=O)CC(N)C(=O)...</chem>	12	0.85	mc-Val-Ala-PABC (0.58)	0.14
immun	<chem>CC(NC(=O)C(NC(=O)CS(=O)(=O)...</chem>	12	0.85	mc-Val-Ala-PABC (0.588)	0.14
immun	<chem>CC(NC(=O)C(NC(=O)C=CC(=O)NC...</chem>	12	0.85	mc-Val-Ala-PABC (0.549)	0.14
immun	<chem>CC(NC(=O)C(NC(=O)CCC(=O)NCC...</chem>	12	0.85	mc-Val-Ala-PABC (0.646)	0.14
immun	<chem>CC(NC(=O)C(NC(=O)C(N)CCC(=O)...</chem>	12	0.85	mc-Val-Ala-PABC (0.563)	0.14
oligo	<chem>NCC1CCC(NC(=O)N2CCN(c3ccc(S...</chem>	6	1.0	MCC (non-cleava (0.145)	0.65
oligo	<chem>NCC1CCC(NC(=O)c2ccc(NC(=O)N...</chem>	6	1.0	MCC (non-cleava (0.158)	0.65
oligo	<chem>NCC1CCC(Nc2ccc(NC(=O)N3CCN(...</chem>	8	1.0	MCC (non-cleava (0.152)	0.4
oligo	<chem>NCC1CCC(NC(=O)c2cnc(-c3cnc(...</chem>	5	1.0	MCC (non-cleava (0.16)	0.85
oligo	<chem>NCC1CCC(NS(=O)(=O)N2CCN(C(=...</chem>	5	1.0	MCC (non-cleava (0.163)	0.85

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